Gabriele Fabbro

Personal Information

Full name: Gabriele Fabbro
Date and place of birth: 10 November 1998, Pordenone (Italy)
Citizenship: Italian
Current job position: Researcher at the Norwegian University of Science and Technology (NTNU)
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Education

Universite' de Toulouse — **Nov. 2022 – 2025**
PhD in Theoretical Chemistry (Adv: Dr. Trond Saue) — *Toulouse, France*

Universita' degli Studi di Trieste — **Oct. 2020 – Oct. 2022**
European Master in Theoretical Chemistry and Computational Modelling [110/110 cum laude] — *Trieste, Italy*

Universita' degli Studi di Trieste — **Oct. 2017 – Sep. 2020**
Bachelor's degree in Chemistry [110/110] — *Trieste, Italy*

Istituto Tecnico Settore Tecnologico J.F.Kennedy — **Sep. 2014 – Aug. 2017**
Pordenone, Italy

Languages

Italian: Native speaker
English: Fluent
French: Basic

Technical Skills

Languages: Fortran 90/95, Python
Text Editors: Microsoft Word, LaTeX
Operating systems: Linux, macOS

Research area

1. Coupled-cluster theory for electronically open systems.
2. Relativistic Coupled-cluster Theory.
3. Highly-accurate determination of nuclear electric quadrupole moments.

Currently, I am part of the **OpenQuantum** project, where I develop coupled-cluster theory for electronically open systems, including molecules and extended systems such as surfaces.
During my PhD, I was a member of the **HAMP-vQED** project, where I focused on the development of highly accurate relativistic coupled-cluster methods. In particular, I developed a diagram-based framework for the automated generation of expectation values at arbitrary excitation rank, using the `tenpi` equation generator. Within this framework, I implemented expectation values at the CCSDT and CCSDTQ levels of theory in the `DIRAC` program package. My research primarily targeted the accurate calculation of the electric field gradient (EFG) at nuclear positions and its application to the reliable extraction of nuclear electric quadrupole moments from molecular systems.

Publications

G. Fabbro and H-M. Høyvik, *The Ensemble Ghost Interaction in Fractional-Occupation Hartree-Fock: Occupation Covariance, Pair Rank, and Prescribed Ensembles of Determinants*, arXiv: https://arxiv.org/abs/2610.38873 *(Corresponding author)*.

G. Fabbro, J. Brandejs, and T. Saue, *The nuclear electric quadrupole moment of ^{87}Sr from highly accurate molecular relativistic calculations*, **J. Phys. Chem. A**, 2026, DOI: 10.1021/acs.jpca.6c00683. *(Selected as cover art)*

G. Fabbro, J. Brandejs, and T. Saue, *Highly Accurate Expectation Values Using High-Order Relativistic Coupled Cluster Theory*, J. Phys. Chem. A, 2025, DOI: 10.1021/acs.jpca.5c02844.

G. Fabbro, J. Pototschnig and T. Saue, *Beyond the Dailey-Townes Model: Chemical Information from the Electric Field Gradient*, J. Phys. Chem. A 2025, *129*, 1006-1025, DOI: 10.1021/acs.jpca.4c06884.

Organization of Workshops/Conferences

1. One of the organizers of 21st ETSF Young Researchers' Meeting | *Trieste, Italy* **May 2025**

Contributions to Electronic Structure Codes

Part of this work involved the development and implementation of coupled-cluster methodologies in existing electronic structure codes.

- `eT`: Implementation of FON-UHF and SA-UHF methods.
- `DIRAC`: Implementation of coupled-cluster expectation values at the CCSDT and CCSDTQ.
- `tenpi`: Implementation of the automated generation of the $\Lambda-$equations and the corresponding coupled-cluster density matrices for arbitrary excitation rank.

Seminars and Schools

Molecular Response Properties Summer School | *Stockholm, Sweden* **June 2023**

Introduction to HPC Computing Systems and Use of the Olympe System | *Toulouse, France* **Jan. 2023**

European Summerschool in Quantum Chemistry (ESQC22) | *Torre Normanna, Italy* **Sep. 2022**

Winter School for advanced sciences of Luchon: Tutorials in Theoretical Chemistry | *Aspet, France* **Feb. 2022**

Advanced Computational Techniques | *Paris, France* **Nov. 2022**

Advanced methods in Electronic Structure, Dynamics and Molecular Modelling | *Toulouse, France* **Sep. 2021**

Scientific oral communications

14th International Conference on REHE | *Amersfoort – The Netherlands* **Oct. 2024**

- Contributed talk: Highly Accurate Molecular Properties using Relativistic Coupled Cluster Theory.

27th DIRAC Working Group Meeting | *Odense, Denmark* **June 2024**

- Scientific talk: Coupled Cluster Method for Molecular Properties.

20th ETSF Young Researchers' Meeting | *Toulouse, France* **May 2024**

- Contributed talk: Coupled Cluster Method for Molecular Properties.

26th DIRAC Working Group Meeting | *Odense, Denmark* **May-June 2023**

- Scientific talk: Analysis and calculation of electric field gradient in relativistic framework.

Molecular Response Properties Summer School | *Stockholm, Sweden* **June 2023**

- Poster session: Beyond the Dailey-Townes model: chemical information from the electric field gradient.

European Summerschool in Quantum Chemistry (ESQC22) | *Torre Normanna, Italy* **Sep. 2022**

- Poster session: Analysis and calculation of electric field gradient in relativistic framework.

Departmental/University service

1. Member of the evaluation committee of M2 TCCM students during their end of Master | *Toulouse, France* **June 2024**